

GEV Hazard Curve Model (MKM-GH-001)

Wolfram Oracle Benchmark of the GEV Implementation

MKM Research Labs

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1 Executive Summary

This document benchmarks the implementation of the GEV Hazard Curve Model (MKM-GH-001): the `GEVFitter` class that fits a generalised extreme value (GEV) distribution to gauge peak levels and turns the fit into return levels and exceedance probabilities. It asks two questions of the code, and answers each against references that share none of its numerics.

1. **Does it compute its formulae correctly?** Return levels and exceedance probabilities are checked against the closed form evaluated to 50 significant digits, and against Wolfram. Within the operating range ($T \leq$ 10,000 years) the worst relative error is 3.0×10^{-11} on a return level and 4.2×10^{-10} on an exceedance probability, against a tolerance of 1.0×10^{-9}. Wolfram and the 50-digit reference agree to 5.0×10^{-30}, so the reference is confirmed by a second, independent implementation.

2. **Does its fit reach the maximum of the likelihood?** The model's maximum-likelihood fit is compared with an independent multi-start optimiser on a likelihood written from the density. The largest gain in log-likelihood the independent fit finds is 2.1×10^{-6} (tolerance 1.0×10^{-4}): the model's optimiser stops at the maximum.

Finding. The implementation is sound; the tail risk is in the *estimate*, not the arithmetic. On samples drawn from known parameters, the fitted 1-in-1000 level ranges from 1.02 to 4.46 times the true one through sampling variation alone (Table 3). The model reports return levels without a confidence interval.

Evidence. 6 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test: `src/models/hazard/gev.py` at commit `8bbb73347e54`.
- SHA-256 of that file: `cca7d20506171a85`...
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-19 by `python -m tools.wolfram_oracle`.

2 Model Purpose and Scope

2.1 What is benchmarked

The four public methods of `GEVFitter` in `src/models/hazard/gev.py`, exactly as the hazard-curve builder calls them: `fit` (GEV, and Gumbel when forced), `return_level`, `exceedance_probability`, and the sign conversion between the extreme-value shape ξ and scipy's $c = -\xi$. The class is loaded from its file alone, so the benchmark exercises the production code and nothing around it.

2.2 What is not

This is a benchmark of the implementation (validation question Q8). It is not independent validation (Q9): it shows the model computes what its formulae say, not that a GEV fitted to per-storm peaks is the right model for the hazard.

3 Mathematical Framework

With location μ, scale $\sigma > 0$ and shape ξ ($\xi > 0$ a heavy Fréchet tail, $\xi < 0$ bounded above), the GEV distribution function is

$$F(x) = \exp\left\{ -\left[1 + \xi \frac{x-\mu}{\sigma}\right]^{-1/\xi} \right\}, \qquad F(x) = \exp\left\{-e^{-(x-\mu)/\sigma}\right\} \quad (\xi = 0). \tag{1}$$

The T-year return level solves $F(x_T) = 1 - 1/T$. Writing the reduced variate $y_T = -\log(1 - 1/T)$,

$$x_T = \mu + \frac{\sigma}{\xi}\left(y_T^{-\xi} - 1\right), \qquad x_T = \mu - \sigma \log y_T \quad (\xi = 0), \tag{2}$$

and the exceedance probability is $P(X > x) = 1 - F(x)$. `GEVFitter` takes the Gumbel branch when $|\xi| < 10^{-10}$.

The log-likelihood of peaks $x_1, \ldots, x_n$, with $t_i = 1 + \xi(x_i - \mu)/\sigma > 0$, is

$$\ell(\mu, \sigma, \xi) = -n \log \sigma - \left(1 + \frac{1}{\xi}\right) \sum_i \log t_i - \sum_i t_i^{-1/\xi}. \tag{3}$$

Sign convention. scipy's `genextreme` uses $c = -\xi$. Passed unflipped, the 1-in-1000 level for $\xi = 0.2$ would be 174.88 rather than 398.07; `GEVFitter` converts in one named place and is correct.

4 Benchmark Method

4.1 Closed-form reference

Each return level and exceedance probability is evaluated with `mpmath` at 50 significant digits, using expm1 and the reduced variate so neither $1 - 1/T$ nor $1 - F$ loses digits. The exceedance threshold for each case is the exact T-year level, so the right answer is $1/T$. Six parameter cases cover the Gumbel, Fréchet, bounded and near-zero-shape regimes, including a shape just below the model's Gumbel switch.

4.2 Wolfram oracle

The same quantities are evaluated in Wolfram with `Quantile` and `SurvivalFunction` of `MaxStableDistribution`$[\mu, \sigma, \xi]$ (same sign of ξ) and, for $\xi = 0$, `ExtremeValueDistribution`. Wolfram's `GumbelDistribution` is the *minimum* law and would be wrong here. Inputs go to Wolfram as the exact value of each double the model used, set to 60-digit precision, so `N[·, 30]` returns 30 genuine digits rather than machine precision.

4.3 Independent fit

The likelihood above is maximised by Nelder–Mead from six starting shapes, with no `scipy.stats` code, and evaluated at 50 digits at both the model's and the independent parameters. Wolfram's `EstimatedDistribution` gives a third fit. A positive gain means the model's optimiser stopped short of the maximum. Data: synthetic samples from known parameters, seed 20261019.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Return levels, $T \leq 10{,}000$	3.0×10^{-11}	1.0×10^{-9}	pass
Exceedance probabilities, $T \leq 10{,}000$	4.2×10^{-10}	1.0×10^{-9}	pass
Wolfram agrees with the reference	5.0×10^{-30}	1.0×10^{-9}	pass
Model agrees with Wolfram	4.2×10^{-10}	1.0×10^{-9}	pass
Fit at the maximum: independent optimiser	2.1×10^{-6}	1.0×10^{-4}	pass
Fit at the maximum: Wolfram	2.1×10^{-6}	1.0×10^{-4}	pass

5.1 Return levels and exceedance

Worst in range: return level at `below_gumbel_switch/T=10000`;
exceedance at `below_gumbel_switch/T=10000`. The `below_gumbel_switch` case carries the model's rounding to Gumbel at $\xi = 10^{-11}$, an error of order ξ that is immaterial. Beyond the operating range the model's $\mathrm{ppf}(1-p)$ and $1 - \mathrm{cdf}$ lose digits as $p \to 0$; this is reported, not judged.

Table 2: GEVFitter against the 50-digit reference: relative errors. The exceedance threshold is the exact T-year level, so $P(X > x_T) = 1/T$

Case	T	Reference level	Level err	Exc. err	Level vs W.	Exc. vs W.
gumbel	10	145.007346546248	1.3×10^{-16}	6.4×10^{-16}	1.7×10^{-16}	6.2×10^{-16}
gumbel	100	192.002984535531	2.3×10^{-17}	6.7×10^{-16}	2.4×10^{-19}	6.8×10^{-16}
gumbel	200	205.916242850700	2.1×10^{-17}	6.8×10^{-16}	8.4×10^{-17}	6.7×10^{-16}
gumbel	1,000	238.145101410474	2.0×10^{-17}	6.4×10^{-16}	6.9×10^{-21}	6.6×10^{-16}
gumbel	10,000	284.205807397854	7.7×10^{-15}	1.1×10^{-13}	7.8×10^{-15}	1.1×10^{-13}
gumbel	1,000,000	376.310201159281	1.5×10^{-12}	2.9×10^{-11}	1.5×10^{-12}	2.9×10^{-11}
frechet_moderate	10	156.842740650253	1.2×10^{-17}	1.6×10^{-16}	4.0×10^{-17}	1.4×10^{-16}
frechet_moderate	100	250.936528171715	2.0×10^{-16}	1.0×10^{-15}	2.1×10^{-16}	1.0×10^{-15}
frechet_moderate	200	288.395445935869	3.9×10^{-17}	6.9×10^{-16}	4.3×10^{-17}	6.8×10^{-16}
frechet_moderate	1,000	398.067345230813	7.2×10^{-17}	5.4×10^{-16}	1.0×10^{-16}	5.5×10^{-16}
frechet_moderate	10,000	630.951034675384	2.2×10^{-14}	1.1×10^{-13}	2.2×10^{-14}	1.1×10^{-13}
frechet_moderate	1,000,000	1584.89303397173	5.8×10^{-12}	2.9×10^{-11}	5.8×10^{-12}	2.9×10^{-11}
frechet_heavy	10	183.231304990444	3.3×10^{-17}	1.3×10^{-16}	1.6×10^{-17}	1.1×10^{-16}
frechet_heavy	100	458.997067605108	5.0×10^{-16}	8.8×10^{-16}	5.3×10^{-16}	9.0×10^{-16}
frechet_heavy	200	624.977284199435	2.8×10^{-16}	1.1×10^{-15}	3.5×10^{-16}	1.1×10^{-15}
frechet_heavy	1,000	1324.59474401879	2.6×10^{-16}	1.1×10^{-15}	3.4×10^{-16}	1.1×10^{-15}
frechet_heavy	10,000	4059.89999708317	5.5×10^{-14}	1.1×10^{-13}	5.5×10^{-14}	1.1×10^{-13}
frechet_heavy	1,000,000	40059.9899999970	1.4×10^{-11}	2.9×10^{-11}	1.4×10^{-11}	2.9×10^{-11}
weibull_bounded	10	136.241869030462	8.7×10^{-17}	4.5×10^{-16}	9.0×10^{-17}	4.2×10^{-16}
weibull_bounded	100	160.149285268037	2.6×10^{-17}	3.6×10^{-16}	4.4×10^{-17}	3.7×10^{-16}
weibull_bounded	200	165.325388660181	4.4×10^{-17}	1.7×10^{-16}	1.9×10^{-17}	1.8×10^{-16}
weibull_bounded	1,000	174.878622625521	5.5×10^{-18}	1.1×10^{-15}	4.8×10^{-18}	1.1×10^{-15}
weibull_bounded	10,000	184.150909578672	2.0×10^{-15}	1.1×10^{-13}	2.0×10^{-15}	1.1×10^{-13}
weibull_bounded	1,000,000	193.690425924240	1.9×10^{-13}	2.9×10^{-11}	1.9×10^{-13}	2.9×10^{-11}
near_zero_shape	10	145.007346596890	4.2×10^{-17}	6.6×10^{-17}	2.2×10^{-17}	8.8×10^{-17}
near_zero_shape	100	192.002984747145	5.1×10^{-17}	4.0×10^{-16}	4.9×10^{-17}	4.1×10^{-16}
near_zero_shape	200	205.916243131156	9.1×10^{-17}	1.4×10^{-15}	9.9×10^{-17}	1.4×10^{-15}

Table 2: GEVFitter against the 50-digit reference: relative errors. The exceedance threshold is the exact T-year level, so $P(X > x_T) = 1/T$

Case	T	Reference level	Level err	Exc. err	Level vs W.	Exc. vs W.
near_zero_shape	1,000	238.145101887576	1.6×10^{-17}	1.1×10^{-15}	1.2×10^{-17}	1.1×10^{-15}
near_zero_shape	10,000	284.205808246148	7.6×10^{-15}	1.1×10^{-13}	7.6×10^{-15}	1.1×10^{-13}
near_zero_shape	1,000,000	376.310203067964	1.5×10^{-12}	2.9×10^{-11}	1.5×10^{-12}	2.9×10^{-11}
below_gumbel_switch	10	145.007346546755	3.5×10^{-12}	2.4×10^{-11}	3.5×10^{-12}	2.4×10^{-11}
below_gumbel_switch	100	192.002984537647	1.1×10^{-11}	1.1×10^{-10}	1.1×10^{-11}	1.1×10^{-10}
below_gumbel_switch	200	205.916242853505	1.4×10^{-11}	1.4×10^{-10}	1.4×10^{-11}	1.4×10^{-10}
below_gumbel_switch	1,000	238.145101415245	2.0×10^{-11}	2.4×10^{-10}	2.0×10^{-11}	2.4×10^{-10}
below_gumbel_switch	10,000	284.205807406337	3.0×10^{-11}	4.2×10^{-10}	3.0×10^{-11}	4.2×10^{-10}
below_gumbel_switch	1,000,000	376.310201178368	5.2×10^{-11}	9.7×10^{-10}	5.2×10^{-11}	9.7×10^{-10}

5.2 Maximum-likelihood fit

Table 3: Maximum-likelihood fit: GEVFitter against an independent optimiser

Series	Fit	n	$\hat{\xi}$ model	$\hat{\xi}$ indep.	ξ true	LL gain	x_{1000} model	x_{1000} true	$\hat{\xi}$ Wolfram	LL gain W.
n30_xi0.1	gev	30	0.2449	0.2449	0.10	1.1×10^{-7}	9.741	6.981	0.2449	1.1×10^{-7}
n30_xi0.1	gumbel	30	0.0000	0.0000	0.10	-4.7×10^{-15}	6.011	6.981	0.0000	-2.0×10^{-14}
n50_xi0.3	gev	50	0.6588	0.6588	0.30	9.0×10^{-8}	54.618	12.256	0.6588	9.0×10^{-8}
n50_xi0.3	gumbel	50	0.0000	0.0000	0.30	-1.8×10^{-15}	7.262	12.256	0.0000	4.2×10^{-14}
n200_xi-0.1	gev	200	-0.0832	-0.0833	-0.10	2.1×10^{-6}	5.087	4.995	-0.0833	2.1×10^{-6}
n200_xi-0.1	gumbel	200	0.0000	0.0000	-0.10	-5.9×10^{-15}	5.659	4.995	0.0000	1.7×10^{-14}
n30_xi0.4	gev	30	0.7073	0.7073	0.40	1.0×10^{-7}	76.048	17.846	0.7073	1.0×10^{-7}
n30_xi0.4	gumbel	30	0.0000	0.0000	0.40	-2.9×10^{-15}	8.996	17.846	0.0000	-2.1×10^{-14}

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The model is shown to compute its formulae; whether those formulae suit the hazard is a separate question.

2. **Per-storm peaks.** `GEVFitter` fits the GEV to every storm's peak level. The GEV is the limiting law for block maxima (e.g. annual maxima); the per-storm choice is a methodology question this benchmark does not answer.

3. **Parameter uncertainty dominates.** At the sample sizes tested, sampling variation moves the 1-in-1000 level by a factor of up to 4.46; the model reports no confidence interval.

4. **Data.** Unless a gauge export is named under Evidence, the fit section uses synthetic samples; it then shows estimator behaviour, not the behaviour on the real gauge record.

7 Document History

Date	Version	Description	Author
19-Oct-2026	0.1	First issue	David K Kelly